

State	Arrangement	Movement	Energy
Solid	Regular lattice, closely packed	Vibrate about fixed positions	Lowest
Liquid	Random, mostly touching	Move/slide over each other	Medium
Gas	Random, far apart	Move rapidly and randomly	Highest

Diffusion: The net movement of particles from a region of higher concentration to lower concentration. Gases with lower relative molecular mass M_r diffuse faster.

2. Atomic Structure: Isotopes

Isotopes are atoms of the same element which have the same number of protons but different numbers of neutrons.

Same Chemical Properties: Because they have the same number of electrons in their outer shell.

Different Physical Properties: Due to different masses (e.g., density, rate of diffusion).

Number of Moles (n):

$$n = \frac{\text{Mass}}{\text{Molar Mass (Ar or Mr)}}$$

Number of Particles:

$$\text{Number of particles} = n \times (6.02 \times 10^{23})$$

Molar Gas Volume:

$$n = \frac{\text{Volume of gas (dm}^3\text{)}}{24} \text{ (at Room Temperature and Pressure)}$$

$$\text{Number of moles } (n) = \text{Concentration } (c) \times \text{Volume } (v)$$

Units for Volume: Must be in dm^3 .

Conversion: To go from cm^3 to dm^3 , divide by 1000.

$$n = c \times \frac{v(\text{cm}^3)}{1000}$$

Units for Concentration: Typically expressed as mol/dm^3 .

Substance Type	Species	Color / Observation	Test / Reaction
Cation	Copper(II)	Blue ppt	Blue ppt with NaOH ; dissolves in excess NH_3 to dark blue solution.
Cation	Iron(II)	Dirty Green ppt	Green ppt with NaOH ; turns brown on standing/surface.
Cation	Iron(III)	Red-Brown ppt	Red-brown ppt with NaOH and NH_3 ; insoluble in excess.
Cation	Chromium(III)	Green ppt	Green ppt with NaOH ; dissolves in excess to dark green solution.
Cation	Zinc (Zn^{2+})	White ppt	White ppt with NaOH and NH_3 ; dissolves in excess (colorless).

Substance Type	Species	Color / Observation	Test / Reaction
Cation	Aluminum	White ppt	White ppt with NaOH (dissolves); insoluble in excess NH_3 .
Cation	Calcium	White ppt	White ppt with NaOH ; no ppt with NH_3 .
Cation	Ammonium	Pungent gas	Warm with NaOH ; gas turns damp red litmus blue .
Anion	Chloride	White ppt	Add acidified Silver Nitrate (AgNO_3); forms AgCl .
Anion	Bromide	Cream ppt	Add acidified Silver Nitrate (AgNO_3); forms AgBr .
Anion	Iodide	Yellow ppt	Add acidified Silver Nitrate (AgNO_3); forms AgI .
Anion	Sulfate	White ppt	Add acidified Barium Nitrate; forms BaSO_4 .
Anion	Sulfite	Purple to Colorless	Add acidified Potassium Manganate(VII).

Substance Type	Species	Color / Observation	Test / Reaction
Anion	Carbonate	Effervescence	Add dilute HCl ; gas turns limewater milky.
Flame Test	Lithium	Red	Clean wire, dip in HCl , then sample, place in flame.
Flame Test	Sodium (	Yellow	Standard flame test procedure.
Flame Test	Potassium	Lilac	Standard flame test procedure.
Flame Test	Copper(II)	Blue-Green	Standard flame test procedure.
Flame Test	Calcium	Orange-Red	"Brick red" flame.
Flame Test	Barium	Apple Green	Standard flame test procedure.
Gas	Chlorine	Pale Green	Bleaches damp litmus paper white.
Gas	Nitrogen Dioxide	Brown	Toxic brown fumes.
Indicator	Methyl Orange	Red / Yellow	Red in acid; Yellow in alkali.

Substance Type	Species	Color / Observation	Test / Reaction
Indicator	Thymolphthalein	Colorless / Blue	Colorless in acid; Blue in alkali.
Solid	Copper(II) Oxide	Black	Basic oxide; reacts with H_2SO_4 to give blue solution.
Solid	Copper(II) Carbonate	Green	Decomposes to black CuO and CO_2 on heating.

Fractional distillation of Crude oil

5. Experimental Techniques: Chromatography

Used to separate mixtures of soluble substances (like inks or drugs).

Baseline: Must be drawn in **pencil** (so it doesn't dissolve and run).

Solvent level: Must be below the baseline.

R_f **Value:** Used to identify substances.

$$R_f = \frac{\text{Distance moved by substance}}{\text{Distance moved by solvent}}$$

Locating Agents: Used to see colorless substances (like amino acids).

Fractional distillation of crude oil

Fraction	Location	Main Function / Use	Boiling Point	Viscosity (Thickness)	Flammability
Refinery Gases	Top	Heating and cooking	Lowest	Very Low (Gas)	Highest
Gasoline (Petrol)	Upper	Fuel for cars	Low	Low (Runny)	High
Naphtha	Upper	Making chemicals/plastics	Medium-Low	Low	Medium-High
Kerosene	Middle	Jet engine fuel	Medium	Medium	Medium
Diesel Oil	Middle	Fuel for trucks and buses	Medium-High	Medium-High	Medium-Low
Fuel Oil	Lower	Fuel for ships and power stations	High	High	Low
Lubricating Oil	Lower	Waxes, polishes, and lubricants	High	High (Viscous)	Very Low

Fraction	Location	Main Function / Use	Boiling Point	Viscosity (Thickness)	Flammability
Bitumen	Bottom	Surfacing roads and roofs	Highest	Extremely High	Lowest

COMPARE BETWEEN COVALENT GIANT STRUCTURE:

Feature	Diamond	Graphite	Silicon(IV) Oxide (SiO ₂)
Bonding	Each C bonded to 4 others	Each C bonded to 3 others	Each Si to 4 O; each O to 2 Si
Structure	Tetrahedral	Hexagonal layers	Similar to Diamond
Hardness	Very hard	Soft/Slippery (layers slide)	Hard
Conductivity	Non-conductor	Conducts (free electrons)	Non-conductor

COMPARING COVALENT, IONIC AND METALLIC BOND

<u>Feature</u>	<u>Ionic Bonding</u>	<u>Simple Covalent</u>	<u>Metallic Bonding</u>
<u>Types of Elements</u>	<u>Metal + Non-metal</u>	<u>Non-metals only</u>	<u>Metals only</u>
<u>Structure</u>	<u>Giant Lattice</u>	<u>Simple Molecular</u>	<u>Giant Lattice</u>
<u>Bonding Force</u>	<u>Electrostatic attraction between oppositely charged ions.</u>	<u>Atoms shared electrons; held by weak intermolecular forces.</u>	<u>Attraction between positive ions and delocalized electrons.</u>
<u>Melting/Boiling Point</u>	<u>High: Strong bonds require high energy to break.</u>	<u>Low: Weak intermolecular forces are easily overcome.</u>	<u>High: Strong metallic bonds throughout the lattice.</u>
<u>Electrical Conductivity</u>	<u>Only when molten or</u>	<u>Non-conductor (no charged</u>	<u>Good conductor</u>

<u>Feature</u>	<u>Ionic Bonding</u>	<u>Simple Covalent</u>	<u>Metallic Bonding</u>
	<u>aqueous</u> <u>(ions are free to move).</u>	<u>particles/free electrons).</u>	<u>(delocalized electrons move).</u>
<u>Physical State at RT</u>	<u>Crystalline Solid</u>	<u>Usually Gas or Liquid</u>	<u>Solid (except Mercury)</u>
<u>Solubility in Water</u>	<u>Usually Soluble</u>	<u>Usually Insoluble</u>	<u>Insoluble</u>

Rate of reaction

1. Concentration (Solutions) or Pressure (Gases)

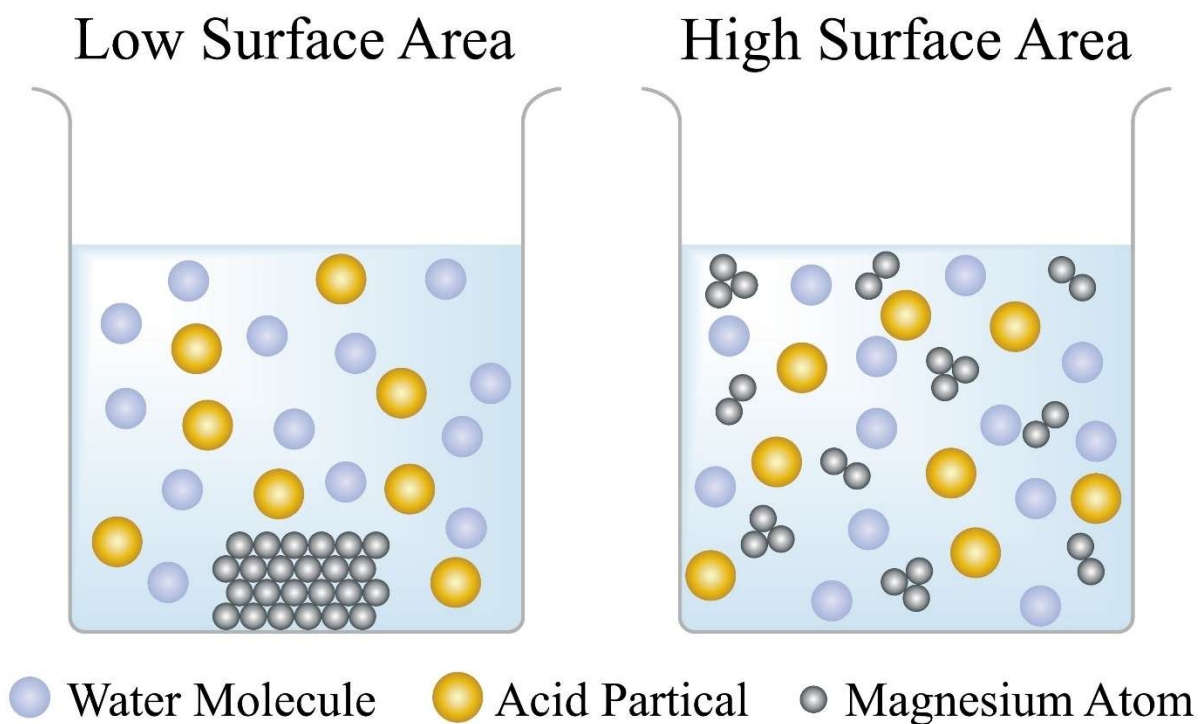
- **The Fact: Increasing concentration or pressure increases the rate.**
- **The Kinetic Explanation: There are more particles in a given volume.**
- **Result: The particles are closer together, leading to more frequent collisions per unit time.**

2. Temperature

- The Fact: Increasing temperature significantly increases the rate.
 - The Kinetic Explanation: Particles gain more kinetic energy and move faster.
 - Result: 1. Particles collide more frequently. 2. More importantly, a much higher proportion of particles have energy greater than or equal to the activation energy (). Therefore, a higher percentage of collisions are successful.
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3. Surface Area (Solids)

- The Fact: Using a powder instead of large lumps increases the rate.
- The Kinetic Explanation: Breaking a solid into smaller pieces increases the total surface area exposed to the other reactant.
- Result: More particles are available to react at the surface at any one time, leading to more frequent collisions.



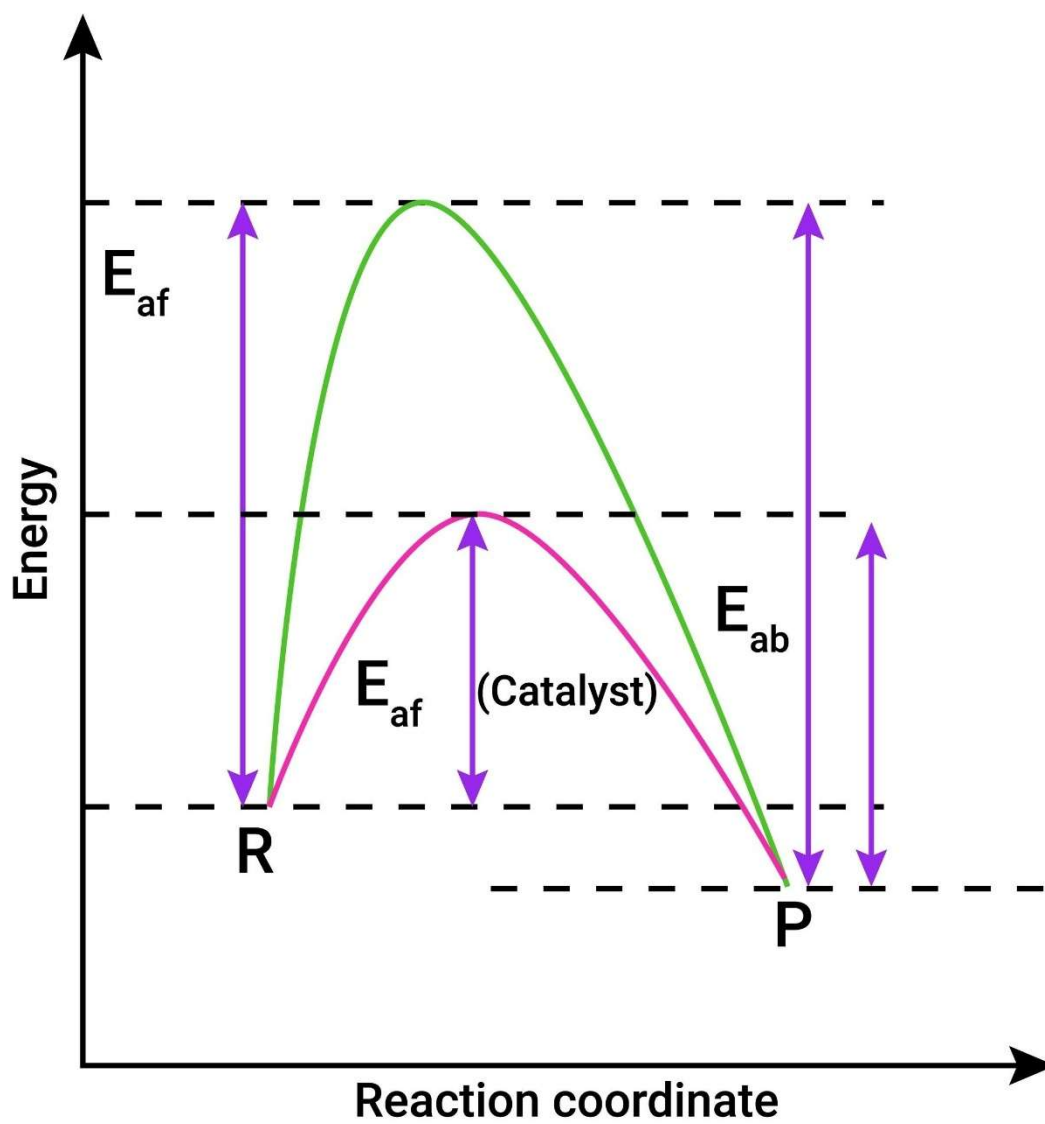
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Explore

4. Catalysts

- **The Fact: Catalysts speed up a reaction without being used up.**
- **The Kinetic Explanation: Catalysts provide an alternative reaction pathway with a lower activation energy.**
- **Result: More particles now have enough energy to react when they collide, increasing the number of successful collisions per unit time.**

Effect of catalyst on energy

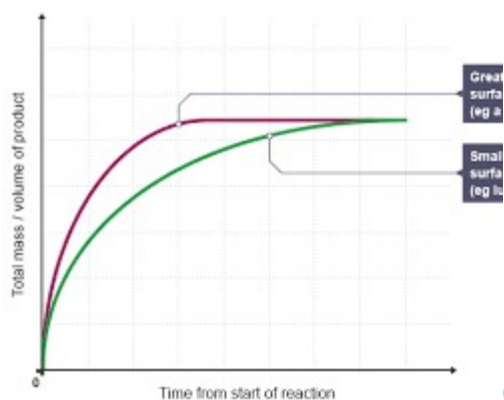


<u>Factor Change</u>	<u>Effect on Gradient</u>	<u>Effect on Time to Finish</u>	<u>Effect on Final Volume</u>
<u>Increase Temp</u>	<u>Steeper</u>	<u>Shorter</u>	<u>Same</u>
<u>Increase Conc.</u>	<u>Steeper</u>	<u>Shorter</u>	<u>Same</u>
<u>Add Catalyst</u>	<u>Steeper</u>	<u>Shorter</u>	<u>Same</u>
<u>Use Powder</u>	<u>Steeper</u>	<u>Shorter</u>	<u>Same</u>
<u>Double Reactant</u>	<u>Steeper</u>	<u>Longer</u>	<u>Double</u>

1. Volume of Gas vs. Time (e.g., \$H_2\$ or \$CO_2\$ production)

This graph shows how much product is made over time.

- **Steeper Gradient: Faster reaction (Higher temperature, higher concentration, or catalyst).**
- **Same Plateau: Same amount of limiting reactant used.**
- **Higher Plateau: More moles of the limiting reactant were added.**



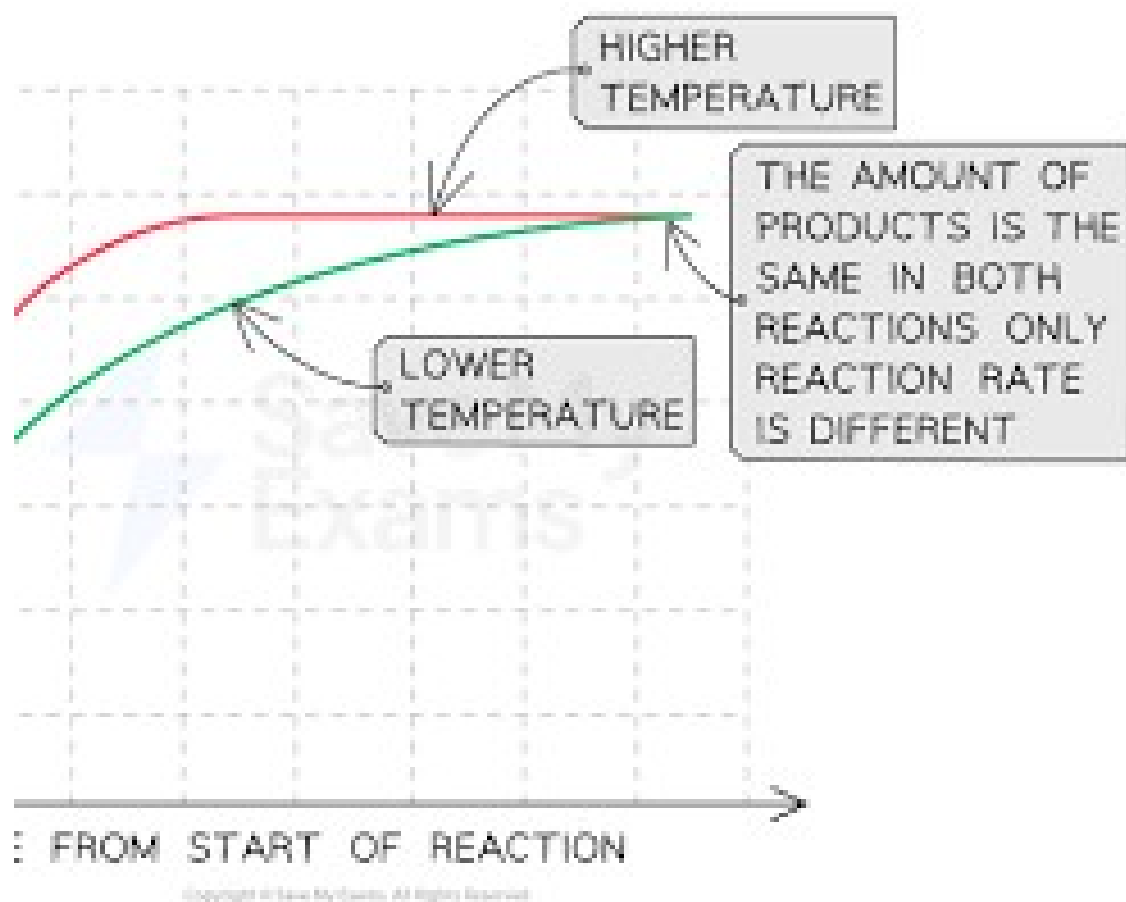
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2. Mass of Reaction vs. Time (Mass Loss)

Used when a gas escapes from a flask. The mass decreases as the gas leaves.

- **Rapid Drop: Faster initial rate.**
- **Horizontal Line: The reaction has finished because a reactant is used up.**

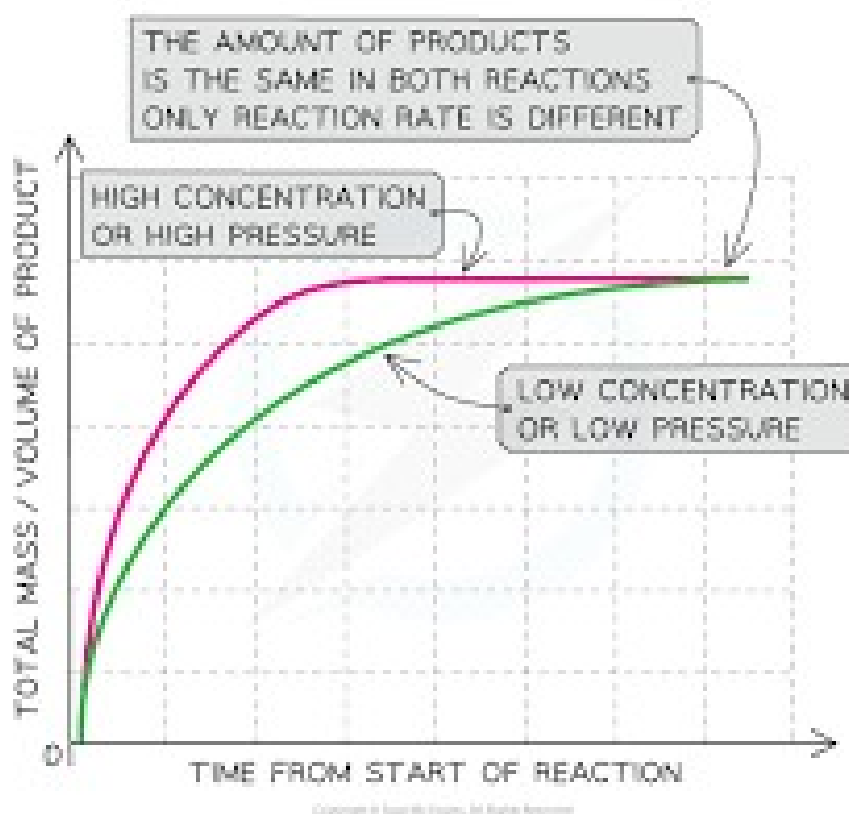


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3. The Effect of Temperature

Increasing temperature increases the kinetic energy of particles.

- The High Temperature curve is much steeper at the start.
- It reaches the finish line faster, but at the same level as the low temperature curve.



In a reversible reaction, the products can react together to reform the reactants. This is represented by the symbol $\rightleftharpoons$. When such a reaction occurs in a closed system (where no substances can enter or leave), it eventually reaches a state called Dynamic Equilibrium.

The Three Conditions for Equilibrium

For a reaction to be at dynamic equilibrium, it must meet these specific criteria:

- 1. Closed System: The reaction must take place in a sealed container so that reactants and products cannot escape.**
 - 2. Equal Rates: The rate of the forward reaction is exactly equal to the rate of the reverse reaction.**
 - 3. Constant Concentrations: The concentrations of reactants and products remain constant (they do not change over time), though they are rarely equal to each other.**
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Le Chatelier's Principle

This principle states that if a change is made to the conditions of a system at equilibrium, the system will shift the position of equilibrium to counteract that change.

1. Effect of Temperature

- Increase Temp: The equilibrium shifts in the direction of the endothermic reaction to absorb the extra heat.**
- Decrease Temp: The equilibrium shifts in the direction of the exothermic reaction to release heat.**

2. Effect of Pressure (Gases only)

- Increase Pressure: The equilibrium shifts to the side with the fewer number of gas molecules to reduce the pressure.**
- Decrease Pressure: The equilibrium shifts to the side with the greater number of gas molecules.**

3. Effect of Concentration

- **Increase Reactant Conc: The equilibrium shifts to the right (products) to use up the added reactant.**
 - **Decrease Product Conc: The equilibrium shifts to the right to replace the removed product.**
-

The Role of a Catalyst

It is a common exam "trap" to ask how a catalyst affects the position of equilibrium.

- **Effect: A catalyst does not change the position of equilibrium and does not increase the yield.**
 - **Reason: It speeds up both the forward and reverse reactions by the same amount.**
 - **Benefit: It allows the system to reach equilibrium faster.**
-

Factor	Change	Shift in Equilibrium Position	Effect on Rate
Temperature	Increase	Shifts in the Endothermic ($\Delta H = +ve$) direction to absorb heat.	Increases
Temperature	Decrease	Shifts in the Exothermic $\Delta H = -ve$) direction to release heat.	Decreases
Pressure (Gases only)	Increase	Shifts to the side with the fewer number of gas molecules.	Increases
Pressure (Gases only)	Decrease	Shifts to the side with the larger number of gas molecules.	Decreases

Factor	Change	Shift in Equilibrium Position	Effect on Rate
Concentration	Increase Reactant	Shifts to the Right (Products) to use up the extra reactant.	Increases
Concentration	Decrease Product	Shifts to the Right (Products) to replace what was removed.	Decreases
Catalyst	Add	No Change (Forward and reverse rates increase equally).	Increases

Polymer Type	Examples	Type of Polymerization	Linkage	Monomers
Polyester	Terylene / PET	Condensation	Ester (-COO-)	Diol + Dicarboxylic acid
Polyamide	Nylon	Condensation	Amide (-CONH-)	Diamine + Dicarboxylic acid
Polyamide	Proteins (Natural)	Condensation	Peptide (-CONH-)	Amino acids
Carbohydrate	Starch (Natural)	Condensation	Ether/Glycosidic (-O-)	Simple sugars (Glucose)
Poly(alkene)	Poly(ethene)	Addition	C-C Single bonds	Alkenes (e.g., ethene)

